

LOSO (glove_ml_explorer) Experiment SB_ypr

Static · YPR only (both hands), no Nothing
 Generated 2026-05-12 00:49:26

Experiment configuration

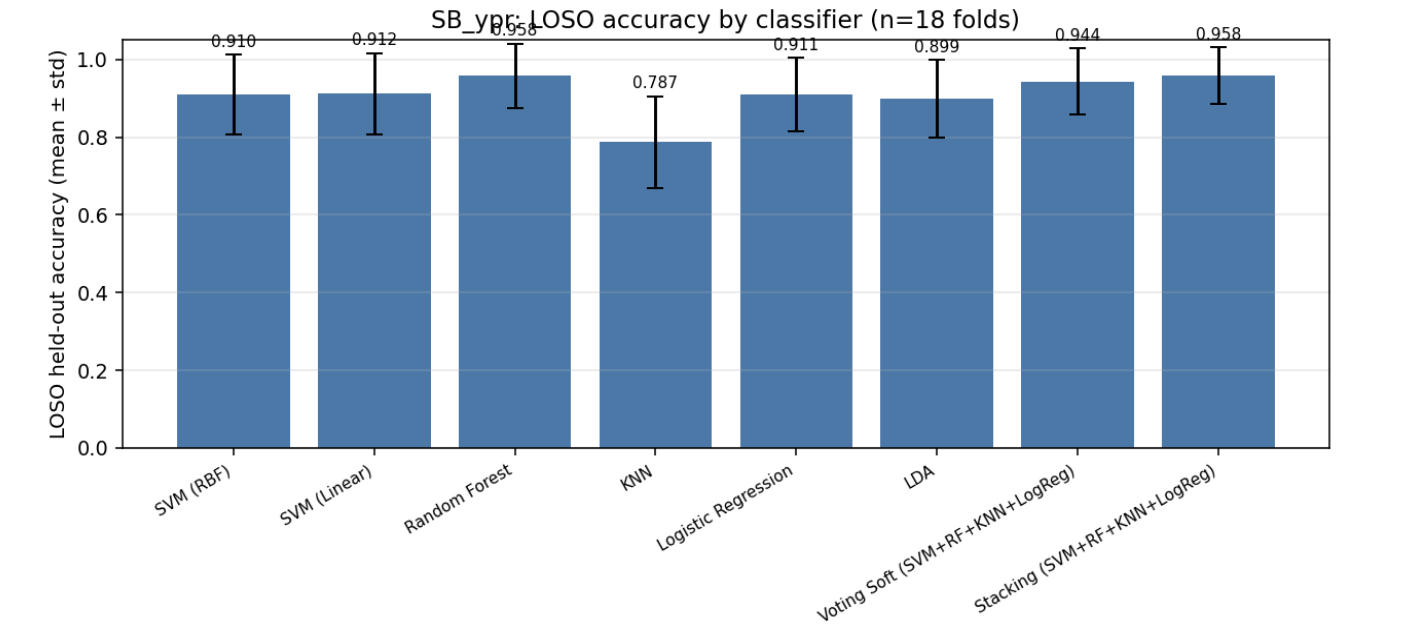
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	YPR
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	66
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	18: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn, Stephen, Val

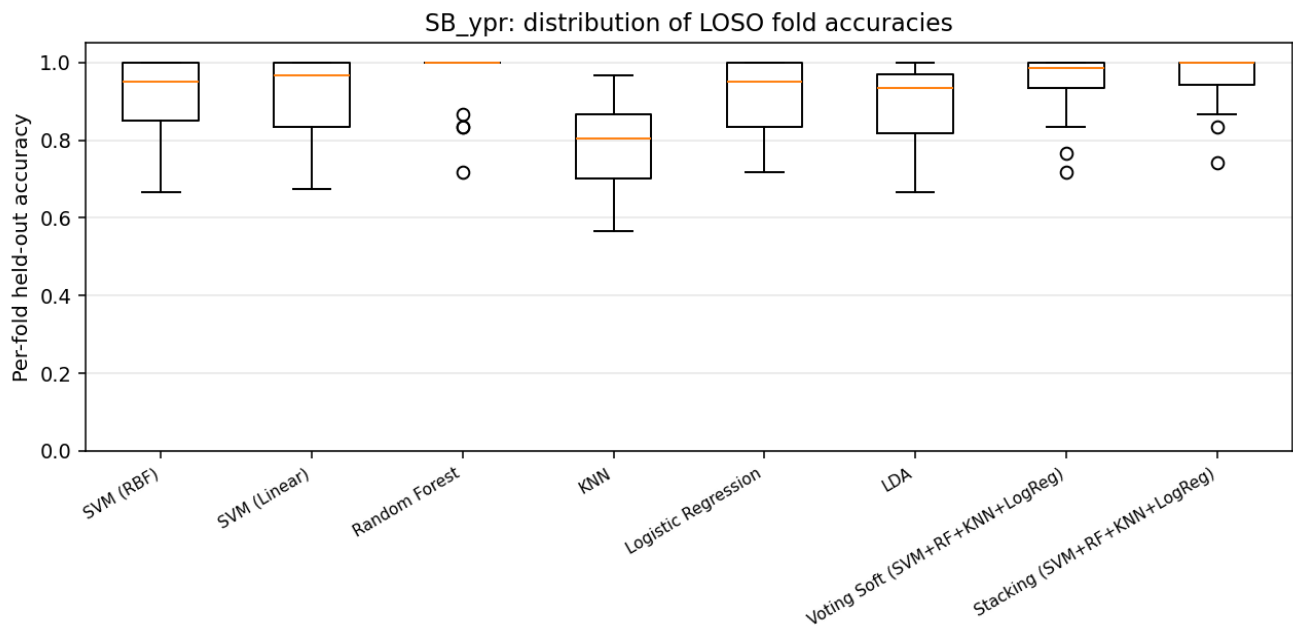
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.9099	0.1020	0.6667	1.0000	0.9781
SVM (Linear)	53,475	0.9120	0.1044	0.6750	1.0000	0.9781
Random Forest	15,468	0.9583	0.0824	0.7167	1.0000	0.9704
KNN	—	0.7874	0.1181	0.5667	0.9667	0.8697
Logistic Regression	21,390	0.9105	0.0941	0.7167	1.0000	0.9716
LDA	21,390	0.8990	0.0996	0.6667	1.0000	0.9278
Voting Soft (SVM+RF+KNN+LogReg)	—	0.9436	0.0841	0.7167	1.0000	0.9818
Stacking (SVM+RF+KNN+LogReg)	—	0.9579	0.0730	0.7417	1.0000	0.9833

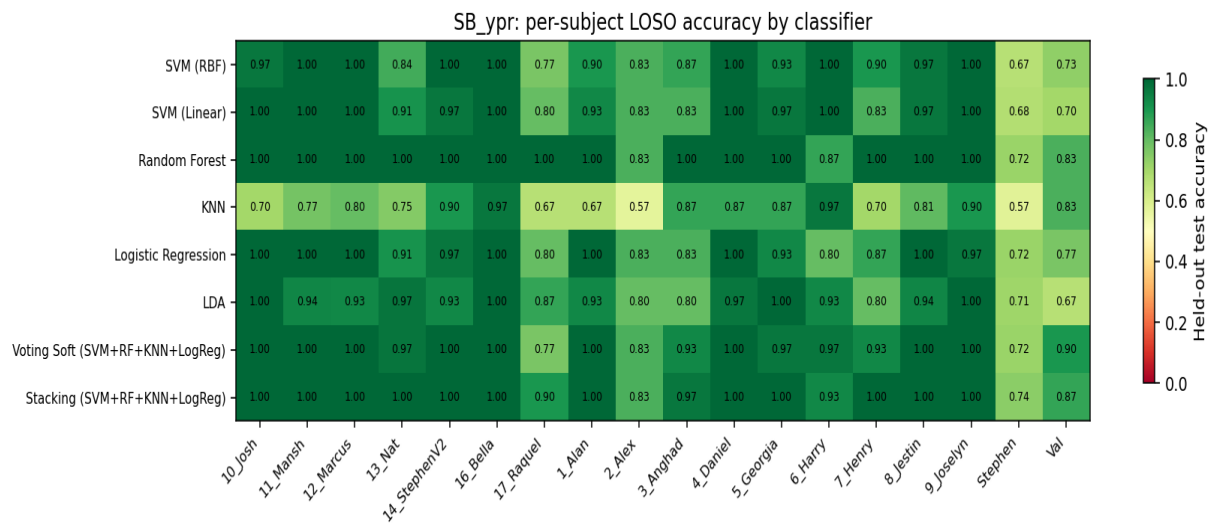
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	0.967	1.000	1.000	0.700	1.000	1.000	1.000	1.000
11_Mansh	603	1.000	1.000	1.000	0.774	1.000	0.935	1.000	1.000
12_Marcus	604	1.000	1.000	1.000	0.800	1.000	0.933	1.000	1.000
13_Nat	602	0.844	0.906	1.000	0.750	0.906	0.969	0.969	1.000
14_StephenV2	604	1.000	0.967	1.000	0.900	0.967	0.933	1.000	1.000
16_Bella	604	1.000	1.000	1.000	0.967	1.000	1.000	1.000	1.000
17_Raquel	604	0.767	0.800	1.000	0.667	0.800	0.867	0.767	0.900
1_Alan	604	0.900	0.933	1.000	0.667	1.000	0.933	1.000	1.000
2_Alex	604	0.833	0.833	0.833	0.567	0.833	0.800	0.833	0.833
3_Anghad	604	0.867	0.833	1.000	0.867	0.833	0.800	0.933	0.967
4_Daniel	604	1.000	1.000	1.000	0.867	1.000	0.967	1.000	1.000
5_Georgia	604	0.933	0.967	1.000	0.867	0.933	1.000	0.967	1.000
6_Harry	604	1.000	1.000	0.867	0.967	0.800	0.933	0.967	0.933
7_Henry	604	0.900	0.833	1.000	0.700	0.867	0.800	0.933	1.000
8_Jestin	603	0.968	0.968	1.000	0.806	1.000	0.935	1.000	1.000
9_Joselyn	604	1.000	1.000	1.000	0.900	0.967	1.000	1.000	1.000
Stephen	514	0.667	0.675	0.717	0.575	0.717	0.708	0.717	0.742
Val	604	0.733	0.700	0.833	0.833	0.767	0.667	0.900	0.867

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.9099 ± 0.1020 (min 0.6667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	4/5	5/5	5/5	0.967
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	1/5	5/6	5/5	0.844
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	0/5	5/5	3/5	5/5	5/5	0.767
1_Alan	5/5	3/5	5/5	4/5	5/5	5/5	0.900
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	2/5	5/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	3/5	5/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	4/5	3/5	5/5	5/5	5/5	0.900
8_Jestin	6/6	5/5	5/5	5/5	4/5	5/5	0.968
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	6/20	15/20	12/20	10/20	19/20	18/20	0.667
Val	3/5	5/5	1/5	4/5	4/5	5/5	0.733
MEAN	0.928	0.875	0.878	0.817	0.966	0.994	0.910

Classifier: SVM (Linear)

Params (best-effort): 53,475

LOSO mean accuracy: 0.9120 ± 0.1044 (min 0.6750, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	2/5	6/6	5/5	0.906
14_StephenV2	4/5	5/5	5/5	5/5	5/5	5/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	1/5	5/5	3/5	5/5	5/5	0.800
1_Alan	5/5	3/5	5/5	5/5	5/5	5/5	0.933
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	3/5	5/5	2/5	5/5	5/5	5/5	0.833
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	3/5	2/5	5/5	5/5	5/5	0.833
8_Jestin	6/6	5/5	5/5	5/5	4/5	5/5	0.968
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	15/20	14/20	11/20	19/20	18/20	0.675
Val	1/5	5/5	1/5	4/5	5/5	5/5	0.700
MEAN	0.878	0.886	0.872	0.853	0.986	0.994	0.912

Classifier: Random Forest

Params (best-effort): 15,468

LOSO mean accuracy: 0.9583 ± 0.0824 (min 0.7167, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	5/5	5/5	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	1/5	5/5	5/5	0.867
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	17/20	19/20	9/20	19/20	18/20	0.717
Val	1/5	5/5	5/5	4/5	5/5	5/5	0.833
MEAN	0.911	0.992	0.997	0.858	0.997	0.994	0.958

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.7874 ± 0.1181 (min 0.5667, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	1/5	3/5	4/5	5/5	5/5	0.700
11_Mansh	3/6	3/5	4/5	4/5	5/5	5/5	0.774
12_Marcus	5/5	3/5	4/5	2/5	5/5	5/5	0.800
13_Nat	6/6	5/5	2/5	0/5	6/6	5/5	0.750
14_StephenV2	5/5	5/5	4/5	3/5	5/5	5/5	0.900
16_Bella	4/5	5/5	5/5	5/5	5/5	5/5	0.967
17_Raquel	5/5	0/5	5/5	0/5	5/5	5/5	0.667
1_Alan	5/5	0/5	5/5	0/5	5/5	5/5	0.667
2_Alex	5/5	2/5	0/5	0/5	5/5	5/5	0.567
3_Anghad	3/5	5/5	4/5	4/5	5/5	5/5	0.867
4_Daniel	5/5	5/5	5/5	1/5	5/5	5/5	0.867
5_Georgia	5/5	1/5	5/5	5/5	5/5	5/5	0.867
6_Harry	5/5	5/5	4/5	5/5	5/5	5/5	0.967
7_Henry	5/5	2/5	0/5	4/5	5/5	5/5	0.700
8_Jestin	6/6	5/5	4/5	0/5	5/5	5/5	0.806
9_Joselyn	5/5	5/5	5/5	2/5	5/5	5/5	0.900
Stephen	3/20	15/20	8/20	7/20	18/20	18/20	0.575
Val	5/5	0/5	5/5	5/5	5/5	5/5	0.833
MEAN	0.869	0.619	0.733	0.508	0.994	0.994	0.787

Classifier: Logistic Regression

Params (best-effort): 21,390

LOSO mean accuracy: 0.9105 ± 0.0941 (min 0.7167, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	4/5	3/5	6/6	5/5	0.906
14_StephenV2	4/5	5/5	5/5	5/5	5/5	5/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	1/5	5/5	3/5	5/5	5/5	0.800
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	3/5	5/5	2/5	5/5	5/5	5/5	0.833
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	3/5	5/5	5/5	5/5	5/5	0.933
6_Harry	5/5	4/5	5/5	1/5	4/5	5/5	0.800
7_Henry	4/5	3/5	4/5	5/5	5/5	5/5	0.867
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	4/5	5/5	5/5	5/5	5/5	5/5	0.967
Stephen	4/20	17/20	13/20	15/20	19/20	18/20	0.717
Val	1/5	5/5	3/5	4/5	5/5	5/5	0.767
MEAN	0.856	0.892	0.903	0.831	0.986	0.994	0.911

Classifier: LDA

Params (best-effort): 21,390

LOSO mean accuracy: 0.8990 ± 0.0996 (min 0.6667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	4/6	5/5	5/5	5/5	5/5	5/5	0.935
12_Marcus	5/5	4/5	5/5	4/5	5/5	5/5	0.933
13_Nat	6/6	5/5	5/5	4/5	6/6	5/5	0.969
14_StephenV2	4/5	5/5	4/5	5/5	5/5	5/5	0.933
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	4/5	5/5	2/5	5/5	5/5	0.867
1_Alan	5/5	4/5	5/5	4/5	5/5	5/5	0.933
2_Alex	4/5	4/5	5/5	1/5	5/5	5/5	0.800
3_Anghad	3/5	5/5	3/5	4/5	5/5	4/5	0.800
4_Daniel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	4/5	4/5	5/5	5/5	0.933
7_Henry	5/5	4/5	4/5	5/5	4/5	2/5	0.800
8_Jestin	5/6	5/5	5/5	4/5	5/5	5/5	0.935
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	15/20	13/20	18/20	17/20	18/20	0.708
Val	1/5	2/5	5/5	4/5	4/5	4/5	0.667
MEAN	0.839	0.897	0.925	0.828	0.969	0.939	0.899

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9436 ± 0.0841 (min 0.7167, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	4/5	6/6	5/5	0.969
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	1/5	5/5	2/5	5/5	5/5	0.767
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	4/5	5/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	4/5	4/5	5/5	5/5	5/5	0.933
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	18/20	14/20	12/20	19/20	18/20	0.717
Val	2/5	5/5	5/5	5/5	5/5	5/5	0.900
MEAN	0.914	0.928	0.961	0.867	0.997	0.994	0.944

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9579 ± 0.0730 (min 0.7417, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	3/5	5/5	4/5	5/5	5/5	0.900
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	4/5	4/5	5/5	0.933
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	18/20	14/20	17/20	18/20	18/20	0.742
Val	2/5	5/5	5/5	5/5	4/5	5/5	0.867
MEAN	0.911	0.972	0.983	0.914	0.972	0.994	0.958